Fiscal Year:	FY 2005	**Task Last Updated:**	FY 02/18/2005
PI Name:	LaCombe, Jeffrey C. Ph.D.		
Project Title:	The Study of Dynamics and Tip Selection in Thermal Dendrites via Pressure Moderated Step Changes in Supercooling		
Division Name:	Physical Sciences		
Program/Discipline:	MATERIALS SCIENCE		
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Glasses and ceramics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	89557-0136	**Congressional District:**	2
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Physical Sciences 01-OBPR-08
Start Date:	11/01/2003	**End Date:**	09/30/2006
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	3	**Monitoring Center:**	NASA MSFC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Koss, Matthew B. (College of the Holy Cross) Glicksman, Martin E. (Rensselaer Polytechnic Institute) Chait, Arnon (NASA Glenn Research Center) Pines, Vladimir (NASA Glenn Research Center)		
Grant/Contract No.:	NNM04AA22G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The objective of this project, titled Tip Selection Dynamics via Pressure Change, is to study fundamental aspects of time-dependent dendritic growth, and obtain benchmark ground-based data on time-dependent and transient dendritic growth phenomena in single, isolated, dendrites. In order to make such measurements, rapid stepwise changes in pressure will be created, which will cause rapid changes in the growth conditions that are not otherwise possible. During these changes, we will closely observe the process in situ, compiling data on the manner in which the process evolves from one steady state to another. The knowledge that this proposed study will produce will improve our understanding of many industrially-important solidification processes by extending the important earlier results of the NASA-sponsored Isothermal Dendritic Growth Experiment (IDGE), by better revealing how dendrites develop over time. This is important in a complex (and difficult to study) environment such as a casting, where boundary conditions change over time. This proposed project grows from an earlier NASA-funded project, the Transient Dendritic Growth Experiment, which successfully passed its Science Concept Review, in preparation for flight aboard the International Space Station. Due to circumstances beyond our control (related to the availability of microgravity resources) we now seek to re-direct our activities towards further developing our understanding of the ground-based process, prior to pursuing a microgravity experiment. The investigator team includes the key personnel of the successful IDGE microgravity experiment, which brings over 40 years of cumulative experience in microgravity experiments. Our experience tells us that maximum return on microgravity resources occurs when the ground-based behavior of the process is understood to the fullest extent possible and the experimental apparatus is developed and tested completely. This proposed project will permit us to move forward in this direction, and continue the long-standing involvement of NASA in key crystal growth scientific studies and technology development.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	During the first year of this grant's activities, we have made good progress in developing a relatively inexpensive and flexible apparatus/system for performing dendritic crystal growth experiments with pressure changes. We developed, procured, and tested a hydraulic pressurization system to induce the pressure changes necessary in our experiments, and we have continued development of a dendritic crystal growth chamber testbed, that will be used with the hydraulic pressurization system. These are described in more detail below. We completed the development of Chamber Testbed A. This chamber uses a combination of commercial off-the-shelf (COTS) components and custom machined parts. Chamber Testbed A was constructed of solid stainless steel flanges in place of transparent windows. It was filled with distilled water only (simplified chamber filling process). It was used to successfully confirm the performance of the pressurization system. In the test, pressure was rapidly changed and there were no leaks in either the chamber or the hydraulic system. A graduate Student Syam Nibhanupudi, was hired in the Fall of 2004 (the earliest opportunity to bring in a new graduate student) and is receiving full support towards his M.S. thesis project as part of this grant. Thus far, he has become conversant in our software, Labview, used for data acquisition and control, and has begun development of Chamber Testbed B. Chamber Testbed B is currently under development. The principal design changes will include the addition of optical windows capable of withstanding the imposed pressure differential. Testbed B will again be tested by filling with distilled water and tested under pressure. Chamber Testbed C is also in the preliminary design phase. The principal design changes will permit testing and experimentation with SCN (rather than water tests). One ampoule of Succinonitrile (SCN) sample material has been received from the purification facility at Marshall Space Flight Center. The Holy Cross team developed software tools for dendritic growth analysis, conducted a study on the tradeoffs between magnification, field of view and measurement precision of dendritic radii. This enables better determination of the necessary optical parameters for dendritic growth experiment set ups. Additionally, a study on the tradeoffs between image compression type and file sizes versus precision of dendritic radii. This enables a better determination of the data transmission and temporary data storage requirements for on orbit dendritic growth experiments.
Bibliography Type:	Description: (Last Updated: 03/10/2006)
Articles in Peer-reviewed Journals	Koss, M.B.; LaCombe, J.C.; Chait, A.; Pines, V.; Zlastkowski, M.; and Glicksman, M.E. "Pressure-Mediated Effects on Thermal Dendrites" To appear in Journal of Crystal Growth , Jan-2005